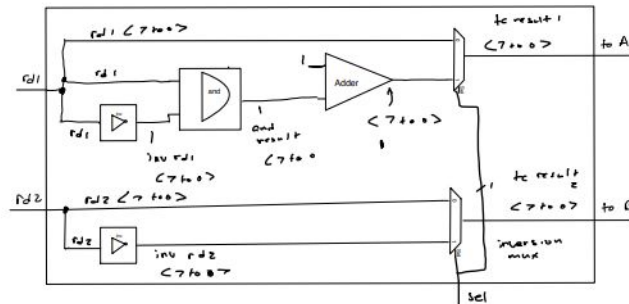
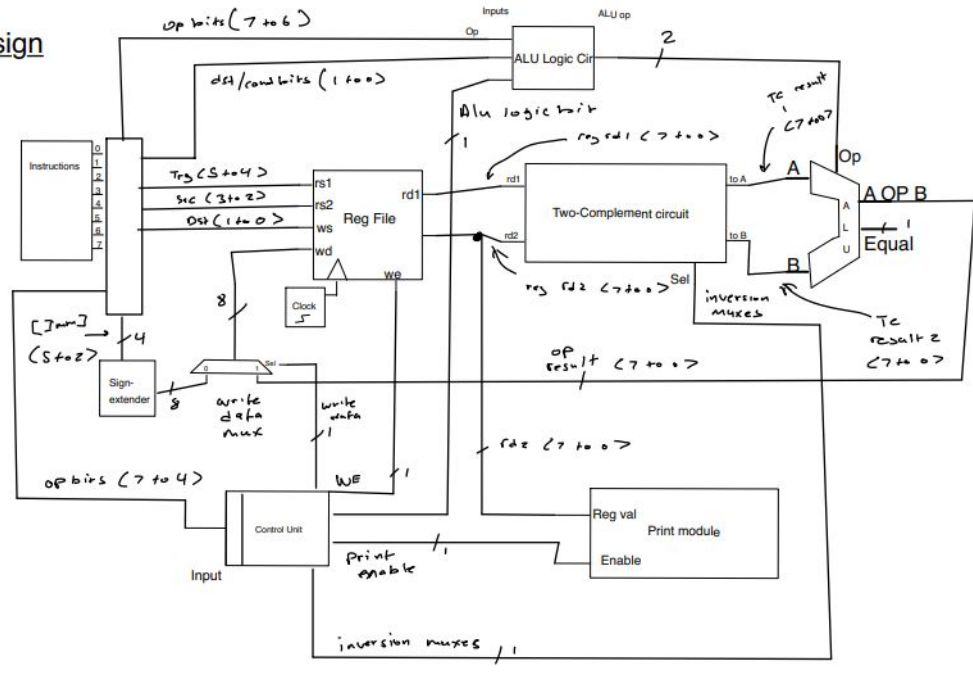

Lab 3: Single cycle calculator in vhdl design

Gerald Stanford • 04.17.20XX

Full Datapath Design



ALU Logic controller TRUTH TABLE

ALU controller logic Truth table										
INPUTS	INPUTS	INPUTS	INPUTS	INPUTS		OUTPUTS	OUTPUTS	FULL OUTPUT	ALU FUNCTION	
op (bit 8)	op (bit 7)	cond bit(bit 2)	cond bit (bit 1)	control unit bit		left bit	right bit			
0	0	0 x	x	0		1	0	10	nothing	
0	0	0 x	x	1		0	0	00-	add	
1	0	0	0	0		1	1	11	set on equal	
1	0	0	0	1		0	1	1	subtract	
1	0	1 X		1		0	0	00-	add	
1	1	1 X	X	0		1	0	10	nothing	
0	1	0	0	1		X	X	X	X	
0	1	0	1	1		X	X	X	X	
0	1	1	0	1		X	X	X	X	
0	1	1	1	1		X	X	X	X	

Why These inputs were chosen

- OP(top 2)
 - Dst/ cond(bottom 2)
 - Alu logic bit
-

Control Unit TRUTH TABLE

Control Unit Truth table									
INPUTS	INPUTS	INPUTS	INPUTS			OUTPUTS	OUTPUTS	OUTPUTS	OUTPUTS
op (bit 8)	op (bit 7)	op (bit 6)	op (bit 5)			we	Write-data mux	inversion muxes	ALU logic bit
0	0	1	0	neg		1	1	1	1
0	0	1	1	print		0	x	x	0
1	0	x	x	add		1	1	0	1
1	1	x	x	load		0	x	x	0